Torsional reinforcement matlab longitudnal Q.2
reinforcement hota hai jiske liye Mue
design karna padta hai, but iss question
me design karne ki zaroorat nahi thi
isiliye MT required nahi tha isiliye Mue
= Mu.

Worksheet 5, a 19. 125mm=bend = 400 D: 425
0.04 x B x D : 0.4 x 250 x 425
1) Astmax : 4% BD : = 4250 mm²

2) Astmin : $\frac{0.85}{f_y}$ Bd

: $\frac{0.85}{415}$ x 250 x 400 : 204.8 m² ≈ 205 mm².

Answer would be option b

note question 4 on the previous page.

Q2.13 The developement length of a deformed
of bar can be expressed as $\frac{1}{k}\frac{\phi \sigma_s}{\tau_{bd}}$. from IS:
456:2000 the value of k can be calculated as →

for plane bar $Ld = \frac{\sigma_s t \phi}{4 \tau_{bd}}$

for deformed bar $Ld = \frac{\sigma_s t \phi}{4 \times 1.6 \times \tau_{bd}} = \frac{\sigma_s t \phi}{6.4 \tau_{bd}}$

⇒ k = 6.4

Q for a beam of width 230 mm & effective depth 500 mm number of bars of 12 mm dia required to satisfy minimum tension reqirema-t specified by IS 456:2000 (assoming grade of Steel Reinforcement Fe 500) is ——— .

Ans width 230 mm, effective depth d = 500 mm

ϕ = 12 mm, minimum tension RF, grade: Fe 500.

n = ?

$$A_{st\,min} = \frac{0.85\,bd}{f_y} = \frac{0.85 \times 230 \times 500}{500} = 195.5\ mm^2$$

$$n = \frac{A_{st\,min}}{\text{Area of 1 bar}} = \frac{195.5\ mm^2}{\frac{\pi}{4} \times 12^2} = \frac{195.5}{113} = 1.7 \simeq 2 \text{ bars}$$

Q. According to recommendation of IS code the compressive of shear not exceed → 4% of CSA.

Q Rf concrete beam is assumed be made of → heterogeneous mat.

		LS of collapse			LS of serviceability		
		DL	LL	WL	DL	LL	WL
①	DL+LL	1.5	1.5	–	1	1	–
②a	DL+WL/EL	1.5	–	1.5	1	–	1
②b	DL will when stab. against OT is crit	0.9*	–	1.5	–	–	1
③	DL+LL+EL/WL	1.2	1.2	1.2	1	0.8	0.8

DL = dead load or self weight

LL = live load or imposed load (cannot act simultaneously)

WL/EL = Wind load / Earthquake .. seismic load

0.9* when stability is overturning is critical.

Coefficient of thermal expansion for aggregate | 94

Limestone : 0.6×10^{-5} to 0.9×10^{-5}/c

QSGBL Pg 16 table.

Quartzite maximum	Limestone minimum.

Column (New Chapter)

Limit state of collapse - compression.

Column is vertical compression member

Main design criteria for column is stability

λ = slenderness ratio = $\frac{l_{eff}}{r_{min}}$

l_{eff} is effective length of column

r_{min} is minimum radius of gyration. or distribution of area

$r_{min} = \sqrt{\frac{I_{min}}{A}}$

other meaning:

$\lambda = \frac{l_{eff}}{ll_d}$

I_{min} = minimum moment of inertia

A is crosssection area.

$3 < \lambda < 12$

↳ long column

Pedestal

Short column.

Codal provisions:

1) Minimum diameter of longitudinal bar provided should not be less than 12 mm.

2) for rectangular column, minimum number of reinforcing bars (longitudnal bars) should not be less than 4.

3) For circular column iminimum number of of bars should not be less than 6.

4) Minimum area of steel reinforcement of column. should not be less than .8%. of gross area/sectional area.

5) Maximum area of reinforcement should not be more than 6% of gross sectional area of column.

is 4% of gross section area if lapping is done. (when bars are lapped)

6) Spacing of longitudinal bars measured along the periphery shall not exceed 300 mm.

7) In case of pedestal in which the longitudnal reinforcement is not taken into account in calculation nominal longitudnal reinforcement should not be less than 0.15% of gross sectional area.

8) In case of helical stirrups/ties, strength or load carrying capacity of column is increased by 5% or a factor of 1.05.

Pedestal is a compression member the effective length of which does not exceed 3 times of its least lateral dimension.

Transverse reinforcement:

transverse reinforcement should be provided in form of lateral ties or lateral stirrups or circular ties or helical stirrups or polygonal links.

Pitch or Spacing of lateral ties

max pitch ≯ least lateral dimension of column mm

≯ 16 x smallest dimension of longitudinal bar or 16 main, min.

≯ 300 mm.

Helical reinforcemen pitch:

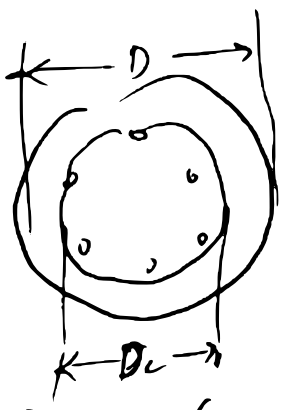
- maximum spacing or maximum pitch ≯ 75mm
  ≯ $\frac{\text{Core diameter of column}}{6}$
  or $\frac{D_c}{6}$
- minimum pitch or minimum spacing ≮ 25mm
  ≮ $3\phi_h$

ϕ_h is diameter of helical bar

Diameter: ① Diameter of lateral ties or polygonal link: minimum diameter of bar should not be less than

or ϕ_{min} ≮ $\frac{\phi_{main,max}}{4}$

≮ 16 mm



① Diameter for helical ties:
above criteria is same for helical ties.
(D is gross dia of column)

Helical ties check. = $\frac{0.36 f_{ck}}{f_y}\left[\frac{A_g}{A_c} - 1\right] \not< \frac{V_h}{V_c}$

$A_g = \pi/4\, D^2$ $A_c = \pi/4\, D_c^2$

V_h is volume of helical ties. & V_c volume of core.

failure pattern of columns.

- pedestal fails in crushing only
- short column - fails in crushing (yielding) as well as in buckling
- long column fails in buckling.

28 March 2023 class 10

Column me main bars kisme dente h

In column main bars or longitudnal bars are in compression.

lateral ties kisme dente h?

- lateral ties or lateral stirrups or helical ties or circular stirrups are always in tension.

Tie kaisa member h?

- Tie – tension member

What is the purpose of lateral ties?

The purpose of lateral ties or lateral stirrups – to ~~provide~~ prevent lateral buckling or buckling of main bars.

RL design and limit state design of column.

assumption basic kyah? strain

① If a column is subjected to axial loading then maximum allowable strain in concrete in axial compression is taken as 0.002.

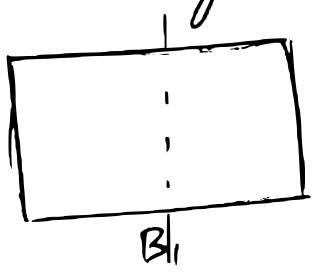
column me outermost compressn fiber pe max compr strain kitna hoga?

② If a column is subjected to axial compression and bending & no tension present in that case maximum compressive strain in outermost compression fiber is $\varepsilon_{max} = 0.0035 - 0.75\,\varepsilon_{min}$

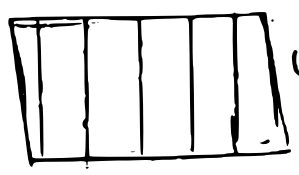
ε_{min} : minimum compressive strain.

How is the strain diagram for statement (1) made?

Strain diagram for statement (1)

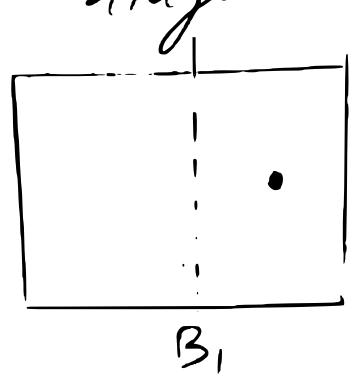


compression only

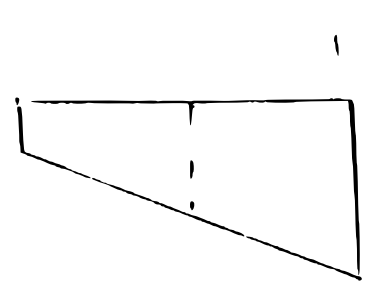


How is the straindiagram for statement (2) made

Strain diagram for statement (2)



AC + Bending [No tension]



Agar min. compressive strain zero h toh max kitna ayega agar column me axial comp ress + bending tension h toh kya hoga?

(3) When column is subjected to axial compressio + bending then there is tension present in that case column behaves like a beam.

minimum eccentricity

All members in compression shall be designed for minimum eccentricity if calculated eccentricity is larger the minimum eccentricity should be ignored

Column and loading conditions

① Short column subjected to axial compression (load)

P_u

② Short column subjected to combined axial compression(load)

P_u, M_{ux} + uniaxial bending

③ Short column subjected to combined axial compression + biaxial bending

P_u, M_{ux}, M_{uy}

Axially loaded Short column

Minimum eccentricity

$e_{min} \geq \frac{l_o}{500} + \frac{LLD}{30}$ ↑

$\geq 20\,mm$

- Saare members ko kisse liye design karte h compression me in?
- calculated ecc ya dash toh ky karenge?
- kya kya conditions h columns ki loading me short columns ke?
- axially loaded short column ki minecentricity kitni hoti
- minecc ka normal/general criteria
- minecc ka max criteria wo $\frac{5}{100}$ LLD wala hai

unsupported length column kya hai?

LLD kya h?

emin kisse bya da nai ho sakta?

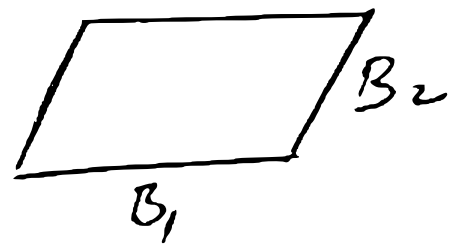
aisa hota h to column kiload carrying capacity kya hogi?

emin me fail hua to kya formula lagayenge?

Pu kya hoga kabuce huga

Puz kya h ka6 use hoga?

l_o is unsupported length of column | 10.2



least lateral dimension B_1 or B_2 ↓

Minimum eccentricity criteria emin.

emin ≯ 0.005 × lateral dimension

emin ≯ 0.05 B_1 . ≯ 0.05 B_2 .

← Condition

if minimum eccentricity follows this criteria then ultimate load carrying capacity of column axially loaded short column shall be calculated by the following equation.

$P_u = 0.4 f_{ck} A_c + 0.67 f_y A_{sc}$

if emin does not follow or does not satisfy the above condition then ultimate load carrying capacity of column can be determined by the following formula

$P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$.

10.5 Puz formulame Kibana % increase hai?

Puz formula is obtained by multiplying 1.11 factor in Pu. ie increasing by 11% in Pu formula.

?, Asc aur Ag kya hai?

Ac is area of concrete
Asc is area of steel in compression
Ag is gross area of column

$A_g = A_c + A_{sc}$.

Helical reinforcement me Pu extra kitni badti hai?

if helical ties are used or helical reinforcement is used then strength increases by 5% or a factor of 1.05 only.

$P_u = 1.05(0.4 f_{ck} A_c + 0.67 f_y A_{sc})$

$P_{uz} = 1.05(0.45 f_{ck} A_c + 0.75 f_y A_{sc})$

Helical ke kaunse grade kisse upar hai ho sakta hai kyon?

f_{ck}: characteristic compressive strength of concrete.

Hook ka max grade kya hai?

f_y: characteristic strength of reinforcement.

Note: Helical reinforcement ke liye maximum grade Fe 415 takose kuna hai.

Member
subjected to
combined
load +
uniaxial
bending
Me kaise
design karenge

Mux Muy
kaise milega
SP16 me
kya hai

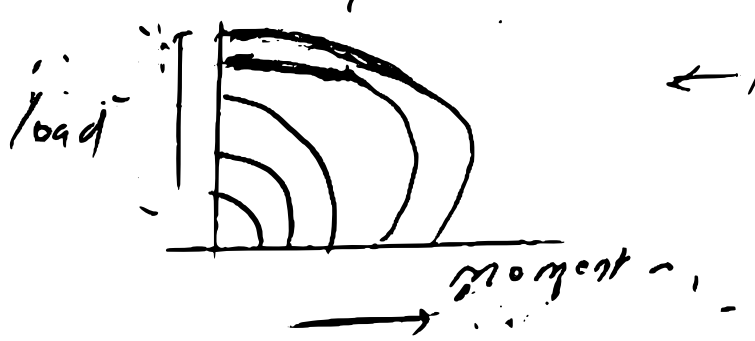
Biaxial
bending
experience
kate colomns
ko kaise design
karenge?

Bressler
2D vs 3D
curve me
kya diff h
kbuse karege?

(B) Member subjected to combined axial load & uniaxial bending

In this case column should be designed for P_{uz}, M_{ux} or M_{uy}

Mux or Muy value is obtained by SP 16. (Bresslers curve or interaction equation)



(C) Members subjected to combined axial load + biaxial bending.

for this case members should be designed for P_{uz}, M_{ux}, M_{uy} ie for designing bending bressles 3D interaction curve is used. (SP-16)

Curve →

Interaction equation 3D ka tyan h?

Interaction equation

$$\left(\frac{M_{ux}}{M_{ux_1}}\right)^{\alpha_n} + \left(\frac{M_{uy}}{M_{uy_1}}\right)^{\alpha_n} \leq 1.0$$

Mux.

Mux, kyah?

Meanings

M_{ux}, M_{uy} = Moments about x & y axis due to design loads.

Interenesr kitnese 3 radani honichahiye?

M_{ux_1}, M_{uy_1} = Maximum uniaxial moment capacity for an axial load of P_u, bending about x & y axes

α_n is related to $\left(\frac{P_u}{P_{uz}}\right)$

Ka ki values kyah?

P_u/P_{uz}	α_n
< 0.2	2
$0.2 - 0.8$	$1+0.2$
> 0.8	2

kispe dep end kotih?

Column me ye above 3 case kis type ke column ke liye applicable h?

$P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$.

Note: point (1)(2)(3) or (A)(B)(C) are applicable for short columns.

Slender column me uniaxial bending aayegi kya?

long column me additional moment ka formula kya h?

lex ley kya h?

Dava b kya h?

lene ka logic kya h?

b kya h?

D kya h?

lo kyon aya h.

long column k. kin kinke liye design karte h?

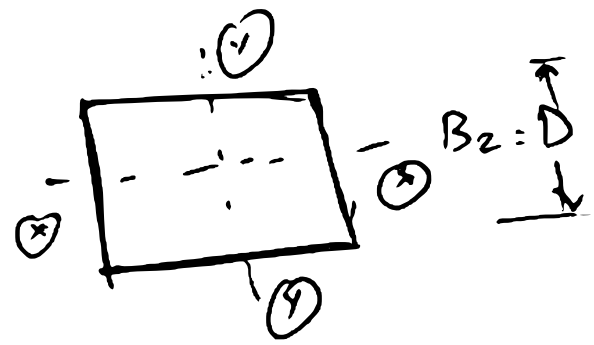
long column or slende-column

long column fails in buckling

In the design of long column, additional moments are used

$$M_{ax} = \frac{P_u D}{2000}\left[\frac{l_{ex}}{D}\right]^2$$

$$M_{ay} = \frac{P_u b}{2000}\left[\frac{l_{ey}}{b}\right]^2$$



l_{ex}, l_{ey} : effective length wrt major & minor axes. or x & y axis.

D : distance ⊥er to major axis

b : distance ⊥er to minor axis

P_u is axial load on member

long column should be designed for $-P_u$

- $M_{ux} + M_{ax}$
- $M_{uy} + M_{ay}$

} Additional moments.

10.9

Short columns ko kaise design karenge.

Short column

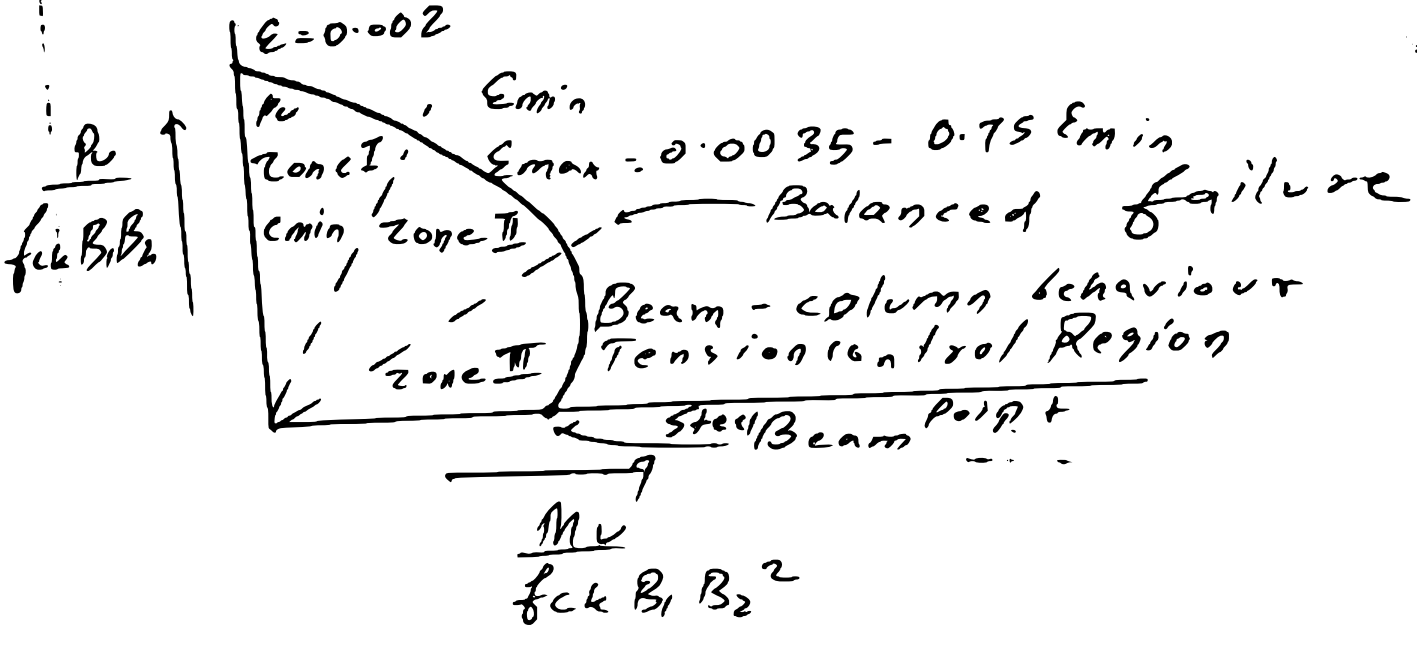
- axial load → e_{min} ✓ → P_u
- axial load → e_{max} → P_{uc}
- axial load + uniaxial bending → P_{u2}, M_{ux} or M_{uy}
- axial load + biaxial bending → P_{u2}, M_{ux}, M_{uy}

long columns ko kiske live design karte h? LSM Me additional moment ko WSM mekya bolte h?

long column → axial load + biaxial bending → P_{u2} · $M_{ux} + M_{ax}$ · $M_{uy} + M_{ay}$

Zone 3 ka strain diagram kaisa hoga?

Understanding Bressler's interaction curve



Zone 1 min eccentricity zone

. In this zone column behaves like axially loaded column and effect of bending is neglected.

name of zone 1 = ε_{min}

. Strain = 0.002

Zone 2 compression control region

. In this zone the effect of load and moment gives equal effect

. Strain ε_{min}

$\varepsilon_{max} = 0.0035 - 0.75\varepsilon_{min}$

Zone 3 Tension control region.

. In this zone the effect of bending is dominating as compared to effect of load hence ~~beam~~ column behaves like beam and this condition is called beam column zone.

Zone III ka strain diagram.

(Strain diagram)

Zone 1 mekya hotah crossler curve?

zone 1 kanaam kyah?

Zone 2 kya uz

kis zone me load ava momt equal effect milta hai?

Zone 3 naam kya hai kya hota hai?

Steam valve kya h

Zone 2 & 3 me

Steel beam pe kyah?

tension control region kyah?

effective length ka definition kya h?

Effective length of column:

effective length is defined as the length between two points of contra flexure (pt of contraflexure woh point h jahaan BM apna sign change karta hai ya jahan BM zero hota hai.

effective length of column depends on kispe depend karta h?

effective length of column (effective) depends on following conditions

1. Type of support
2. Sway or nonsway condition
3. Braced or unbraced condition.
4. unsupported length of column

S C B UB UB

l_{eff} kisse independent h?

effective length of column is independent of span of beam as well as loading on beam.

summary

Design of compression member in WSM

WSM

i) Pedestal & short column with lateral tie

Pedestal short column with lateral tie

Permissible axial load

$P = \sigma_{cc} A_c + \sigma_{sc} A_{sc}$

σ_{cc} is permissible stress in concrete in direct compression.

A_c is Area of concrete.

σ_{sc} is permissible compressive stress of steel

A_{sc} is cross sectional area of longitudinal steel

σ_{cc} = Grade of Concrete — Direct (N/mm^2) compressive stress of concrete σ_{cc}

Grade of Concrete	Direct (N/mm^2) compressive stress of concrete σ_{cc}
M 10	2.5 (3x 4)
M 15	
M 20	5
M 25	6 (7x 8)
M 30	
M 35	9
M 40	10
M 45	11
M 50	12

σ_{sc} for mild steel = Fe 250 = 130 N/mm^2

HYSD bars = Fe 415 = 190 N/mm^2

Fe 500 = 190 N/mm^2

Short column with helical reinforcement
- the above value of P_u increased by 5% (1.05 factor) if helical reinforcement is used

short column ke type P kitna increase krasenge Helical of ke calcme.

WSM

long column or slender column.

long column mein kya hoga?

Permissible axial load

$P = C_r [\sigma_{cc} A_c + \sigma_{sc} A_{sc}]$

C_r kya h?

C_r is reduction coefficient.

reduction coeff ka formula kya h?

1) $\lambda = 12$ to 40

$$C_r = \left[1.25 - \frac{l_{eff}}{48 LLD}\right]$$

dangerous kya hota ke?

2) $\lambda > 40$

$$C_r = \left[1.25 - \frac{l_{eff}}{160 \times r_{min}}\right]$$

λ: slenderness ratio kya hoga

C_r ke formulae me kya lagega?

r_{min} is minimum radius of gyration.

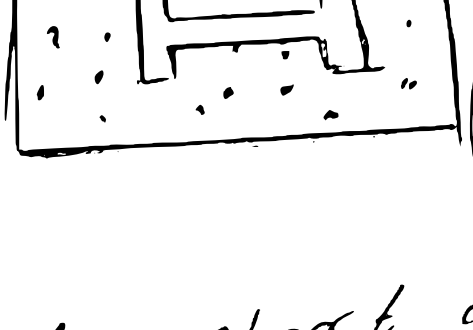
r_{min} ka formula kya h?

$$r_{min} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{\text{Minimum moment of inertia}}{\text{Area of cross section}}}$$

composite column kya hota hai?

Composite column

- When column consists of steel or cast iron or metal core along with RCC then it is called composite construction.



short composite column me formula kya hoga?

$P = (\sigma_{cc} A_{sc} + \sigma_{sc} A_{sc} + \sigma_{mc} A_m)$

Permissible axial load.

- for short composite column.

long composite column me kya changes hoga formula?

- for long composite column

$P = C_r[\sigma_{cc} A_c + \sigma_{sc} A_s] + \sigma_{mc} A_m$

σ_{mc} & A_m are = allowable unit stress of in metal core.

σ_{mc} kya hai?

- for steel core $\sigma_{mc} = 125 N/mm^2$

steel σ_{mc} = ?

- for cast iron core $\sigma_{mc} = 70 N/mm^2$

cast iron σ_{mc} = ?

A_m area of metal core

A_m kya hai?

A_g or gross area = $A_c + A_m + A_{sc}$

gross area kya hoga?

- Note: $A_m \neq 20\% A_g$

Area of metal core kitna hona chahiye?

Area of metal core should not be more than 20% of gross area $A_m \neq 20\% A_g$

11.9)

WSM ke interacn eqn me kya kya aata h?

Kaunsido condns pe hase Ghe design karte?

kya hota h interacn eqn ? formula

$\sigma_{cc,cal}$
$\sigma_{cbc,cal}$
σ_{cc}
σ_{cbc}
kya h?

σ_{cbc} kaise nikalenge?

M20 ka σ_{cbc} kya h?

M25 ka σ_{cbc} kya h?

WSM

Members subjected to combined axial load & bending.

1) Design based on uncracked section

A member subjected to axial load and bending shall be considered safe provided the following conditions are satisfied:

$$\frac{\sigma_{cc,cal}}{\sigma_{cc}} + \frac{\sigma_{cbc,cal}}{\sigma_{cbc}} \leq 1$$

$\sigma_{cc,cal}$ = calculated direct compressive stress in concrete

σ_{cc} = permissible direct compressive stress in concrete.

$\sigma_{cbc,cal}$ = calculated bending compressive stress in concrete

σ_{cbc} = permissible bending compressive stress in concrete.

$$\sigma_{cbc} = \frac{f_{ck}}{3}$$

σ_{cbc} for M20 : 7 N/mm²

σ_{cbc} for M25 = 8.5 N/mm²

final deflection including xx kitu ana chahiye ya kitne sezyada nitona chahiye finals:

partitions ke karan mars kitna?

WSM ke hisab se concrete me kause do type ke cracks?

What are in general deflection criteria?

- Deflection.

Codal provisions

(1) The final deflection due to all loads including the effect of temperature, creep & shrinkage & measured from as-cast level of supports of floor, roof and all other horizontal members

$\pm \frac{span}{250}$ or more $\frac{span}{250}$.

(2) The deflection including the effects of temperature, creep and shrinkage after erection of partitions & the application of finishes $\pm \frac{span}{350}$ or more $\frac{span}{350}$

$\pm$ 20mm

12.1 30 March 2024 Class 12

What are deflection criteria
- vertical limit? How is it defined
- Beams me (cantilever kitne kyun?
- Simply Supp?
- Continuous?
- kahan tak valid hai?
- ye effective depth ya total depth hai?
- love upar kaise karenge?
- ye kiske liye valid nahi hai?

Deflection criteria as per IS 456
Vertical deflection limit. span to depth ratio limit
Deflection is defined by ^
Basic values of span to effective depth ratio ($\frac{l}{d}$) for beam as well as one way slab. for span upto 10m.

		A
(1)	cantilever	7
(2)	Simply supported	20
(3)	Continuous	26

for members of span more than 10m span to effective depth ratio should be multiplied by ie A value $\frac{10m}{span\,m}$ but this modification is not valid for cantilever

Q.1 Given A beam of 10m span simply supported deflection criteria for beam is — ? and also find the value of effective depth for the given beam.

① Ans for simply supp

Span l : 10 m

$\frac{l}{d}$: (A) value : 20

$\frac{l}{d} : 20 \Rightarrow \frac{l}{20} : d$

$\Rightarrow \frac{10m}{20} : d$

$\Rightarrow d : 0.5m$ or $500\ mm$

Q2 In the above question if beam span is 12 m then find out the same values.

② Ans: $l = 12\ m \Rightarrow$ it is more than 10 m

modification :-

$: (A)\ value \times \frac{10\ m}{span\ m}$

$: 20 \times \frac{10}{12} = 16.67.$

$\frac{l}{d} = 16.67 \Rightarrow d = \frac{l}{16.67} = \frac{12m}{16.67}$

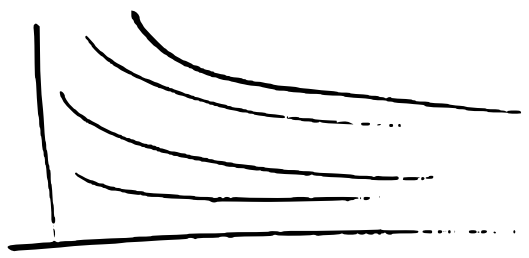
$\Rightarrow d : 0.72m$ or $720\ mm.$

12.3

Deflection for slabs (two way) kaise defined hai

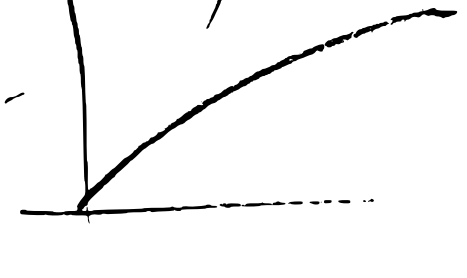
Compression reinforcement ke case me multiplying factor se bane graph ka pattern /trend kaisa hai.

3. if the stress in steel at service load is given for different grades then modification factor should be multiplied based on graph given in Is 456.



← pattern decreasing

4. if compression reinforcement is used then modification is done by multiplying factor given in graph as per Is 456.



← pattern increasing

Deflection criteria for 2way Slab - for slab having shorter span upto 3.5 metre and loading class upto 3 kN/m²

l/D ss slab

twowayMs

ke liye kitna

hota h?

l/D continuous ~~ss~~ slab

two way ke

liye MS

ke liyekya

hota h?

case

kisme

2yada

depth of

steel

mil rha h?

Deflection criteria is defined based on span to overall depth ratio. $\left(\frac{l}{D}\right)$ 12.4

Condition	l/D	l/D
	Mildsteel Fe 250	HYSD bars Fe 415
(1) S/S two way slab	35	35×0.8 = 28
(2) Continuous two way slab	40	40×0.8 = 32

Note: if HYSD bars are used then ~~ge~~ total depth of beam ~~of~~ obtained is more as compared to mild steel.

IS of serviceability - Cracking

condition	maximum crack width
(1) In general	- 0.3 mm
(2) Aggressive envmt or exposed to moisture	0.2 mm
(3) (Severe) aggresiveen vircment (severe)	0.1 mm

limit state of serviceability - corrosion

To prevent or to protect reinforcement from corrosion protective sleeve, ie paint, can be applied on the surface of reinforcing bars.

fire ke zokre ke liye asbestos fet lagate hain ya ocinforie me asbestos paint krte h.

haomful envizonmt me naih toh kitna Crack width desktein?

Max crack width moisture me kitna lenge? Severe me kitna lenge?

corrosion and fire protection ke kya ways h?

limit state of service 12.6

ability - vibration.

Codal provisions:

Water used for RCC should not have pH value less than 6

- No seawater or creek water is used

IS 269 : OPC ka specifin

IS 875 (part 1): Dead load

IS 875 (part 2): imposed load (live load)

IS 875 (part 3): Wind load

IS 875 (part 4): Snow loads

IS 875 (part 5): special load & load combinatio

RCC ke liye pani ka pH kitna lenge.

Nale ke pani se RCC banaskte h kinai?

OPC ka code no kya h?

dead load ka code kya h?

875 part 2 kya h?

Is 875 ka kaunsa part wind load h?

Is 875 part 5?

loadks ka is code kaunsa hai.

2062 kiska code kz

IS 1893 kiske related kz

Dobson unit se kya naapte hi?

456 kisha code kz

Deep beam kab hoti kz

IS 2062: Steel for general structural purpose

IS 1893: criteria for earthquake resistant design of structure /seismic effect

eg to Richter scale pe mark kote hi

Ozone thickness dobson unit

456: PCC & RCC design code of practice.

Deep beam.
A beam shall be deemed to be a deep beam when the ratio of effective span to overall depth is less than

Deep beam ko
kis chiz ke
liye design
karte hai

Sf ke liye
kya karte
hai

(1) $\frac{l_{eff}}{D} < 2$ for SSB 12.8

(2) $\frac{l_{eff}}{D} < 2.5$ for continuous beam

Deep beams are designed for bending moments and checked for shear...

12.9)

Slab kiske liye design krte h?

Design criteria kya hoga a slab k liye?

Aspect ratio kya hota h?
Doosra name kya h?

Oneway slab ka aspect ratio kya hota h?

two way slab kisi edge ke along ko kaise support krte h?

one way ko kaise?

what is deflection a function of? in slabs?

(New Chap) Slab

- Slab is very safe in shear & bending. Hence the main checking criteria for slab is deflection.

- Design criteria for slab is based on loading ie bending moment, shear force, torsion.

- Aspect ratio is $\frac{\text{longer span}}{\text{shorter span}} = \frac{l_y}{l_x}$

- Aspect ratio is also called dimension ratio

One way slab

(i) $\frac{l_y}{l_x} > 2$

(ii) Supported generally along longer edges

(iii) deflection or design is primarily function of shorter span only

Two way slab

(i) $\frac{l_y}{l_x} \leq 2$

(ii) supported generally along all edges.

(iii) Same as one way

slab me load kaun letah?

Distribution bars ka kya use h?

load ko uniformly kaun failatah?

Nominal cover kitna detch slab mein.

kam se kam kitna area of denge slab me

. In slab function of main bar is to take load effectively

. The purpose of distribution bars in slab -

. distribution bars are provided to take care of secondary stresses such as

. temperature (thermal stress)

. creep stress

. shrinkage stress

. distribution bars are provided to distribute the load uniformly

12.10

Codal Provisions:

. Nominal cover provided in slab is 20 mm or 25 mm.

. Minimum area of reinforcement provided in slab

. for Mild steel ($A_{st,min}$)

12.11 |

Hysd bar ke Laseme total area ka kitna denge?

0.15% mallab kya factor hoga?

kiske case me kam area denge steel ka?

Zyada se Zyada main bars kitna door rakhenge?

d kya h slab me?

distribution bars ke liye spacing criteria kya h?

Astmin ✗ 0.15% of Ag

- for HYSD bar

Astmin ✗ 0.12% of Ag

Ag is gross area or gross sectional area

- Note: that means if HYSD bar is used then less area of steel can be provided.

- Maximum spacing criteria (Smax)

for main bars

∴ Smax ✗ 3d

✗ 300mm

- d is effective thickness of slab

for distribution bars

∴ Smax ✗ 5d

✗ 450mm

badese bada kitna dia daal skte h baar ka slab me.

Round off maximum ke case my 2yada se 2yada ke case me kis side karenge.

- Maximum diameter of bar in slab.
- $\phi_{max} \not\propto \frac{D}{8}$ or $\frac{\text{thickness of slab}}{8}$

- Note: always round off the diameter value @ lower side
- given thickness of slab is 250 mm then what is the maximum diameter of bar that can be provided in slab.
- Ans: $\phi_{max} \propto \frac{D}{8} = \frac{250}{8} = 31.25$

$= 31.25\,m$

30 mm ~~32 mm~~

round off to lower side.

12.13/ bade se bada size aggregate ka kitna leng?

flat slab kya hota h?

kis slab ko design nai kaha na padta?

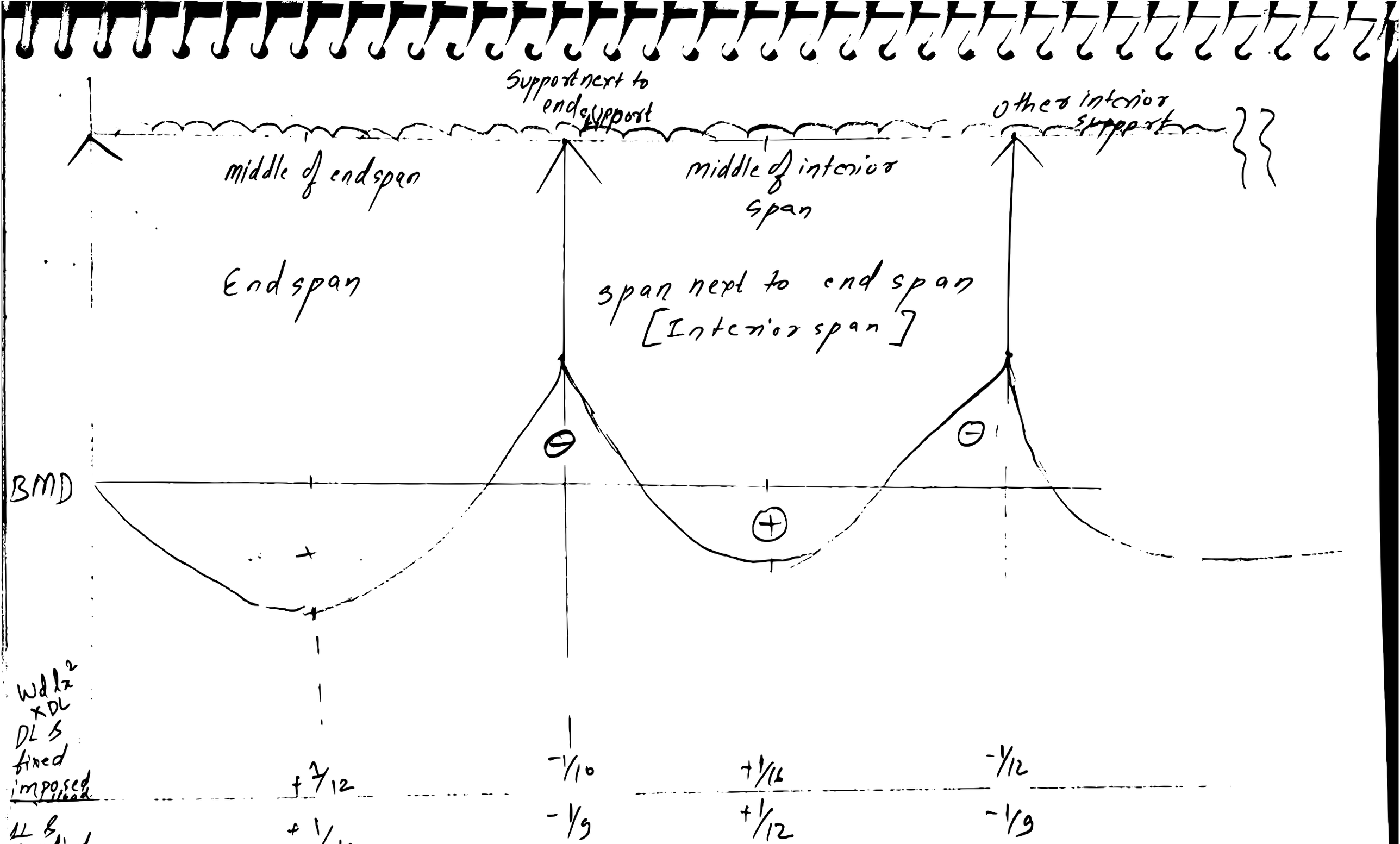
kya kuch fix imposed load?

- Maximum size of Aggregate used in slab

$\frac{1}{4} D$ or $\frac{\text{thickness of slab}}{4}$

- should not be more than this matbal lower sided pe round off karenge

Bending moment coefficients for slab. code mein page 36.

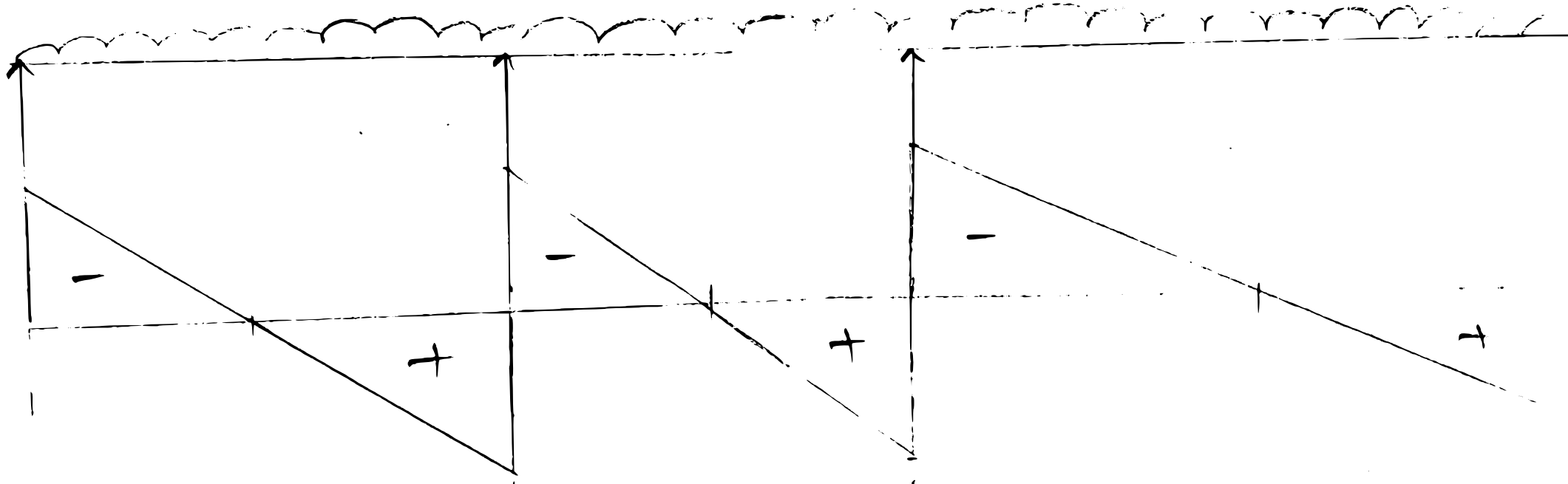

Support next to end support
other interior support
middle of end span
middle of interior span
End span
span next to end span
[Interior span]
BMD
+
−
−
+
−
$Wd l_x^2$ x DL
DL & fixed imposed
+7/12
−1/10
+1/16
−1/12
LL & non fixed imposed load
+1/10
−1/9
+1/12
−1/9
$Wl l_x^2$ x LL

12.14

31 March 2025

Class 13

SFD



0.4 | 0.6 | 0.55 | 0.5

0.45 | 0.6 | 0.6 | 0.6

Wd lx x DL

Wl lx x LL

13.1

Slab design ke
kya methods
h one way
slab me (kis theory pe based)
Design coeff
kaise nikalenge? (kis theory pe based)

two way slab
me kya condns
h (design kis method pe based h)
corner os free
to lift me
kya kya method
h?

marcuss
pig, is method
kya cheez ke
methods h?

One way slab
concentrated
load wale ko
kaise design
karenge?

Design methods for slab.

(1) One way slab - analysis based on elastic theory.

Design coefficients are based on westergaards patteon loadings

Note: One way slab is designed as one way continuous slab.

One way slab with concentrated load should be designed based on

(2) Two way slab — Pigeauds moment coefficient

Two way slab is designed based on

1) When corners not held down or when corners are free to lift. - Rankine Grasshoffs method

2) When corners are held down or corners not free to lift
   1 Marcus correction
   2 pigeauds method
   3 IS code method.

13.3

Rankine grasshoff method kya h?

isme load in x aur y dir me load kaise nikalenge

& kya h?

x dir me moments

y dir me moments kya honge

one way slab se kaise diff h

Marcus correction method kya h?

jo pure wale Rankine se kaise diff h

C kya h?

Rankine grasshoff Method (Timoshenko)

Total load on slab = $W_x + W_y$ (total load sum in x aur y direction ka)

load in x direction $W_x = w \left(\frac{s^4}{1+s^4} \right)$

load in y direction $W_y = w \left[\frac{1}{1+s^4} \right]$

s is aspect ratio $\frac{l_y}{l_x}$ or $\frac{\text{longer span}}{\text{shorter span}}$

$M_x = W_x \frac{l_x^2}{8}$

$M_y = W_y \frac{l_y^2}{8}$

} In general x dir aur y dir me moments

Marcus correction method.

$W_x = w \frac{s^4}{1+s^4}$

$W_y = w \left(\frac{1}{1+s^4} \right)$

$M_x = C * \frac{W_x l_x^2}{8}$

$M_y = C * \frac{W_y l_y^2}{8}$

$C = 1 - \frac{5}{6} \frac{s^2}{1+s^4}$

C ka range/limit kya h?

Pigeauds method me kya kya (radoorthi) chiz h? kya alag h isme/Marcus wale method se?

Cx Cy kya h?

IS code method me kya kya hota h?

Is code ne kyo mana ki moment shorter span ka shi function h?

Wr kya h?

α kya h?

Kis theory pe IS code based h

α kis pr dep end krta h?

C is marcus correction

∴ C cannot be more than 1

Pigeauds' method.

$$M_x = C_x * \frac{w l_x^2}{8}$$

$$M_y = C_y * \frac{w l_y^2}{8}$$

C_x and C_y are corrections based on support condition.

IS code method

$$M_x = \alpha_x w \boxed{l_x^2}$$

$$M_y = \alpha_y w \boxed{l_x^2}$$

w = total load

α_x α_y = moment coefficient based on support condition given in IS 456 which is based on johnsons' yield line theory.

13.5) x dir me SF kitna hoga y dir me?

kab kab torsion of deye?

jab edges continuous h tab deye kya torsion reinforcement?

Do no edges discontinuous me deye kya?

kya analogy? (kharcha gaye parent ke saath vs Akele gaye kharcha)

kitna deye? Ast Torsion aur Ast me kya relation?

Shear force values.

$S_x = \frac{1}{3} w l_x$

$S_y = w l_x \left(\frac{x}{x+2}\right)$

Torsion reinforcement condition:

1. When corners held down (when edges are continuous)
- when edges are continuous then corner is not free for lift hence no torsion occurs hence no torsion reinforcement is required or provided $A_{st} = Nil$

2. When both edges or two edges discontinuous - torsion occurs @ corner hence - torsion reinforcement is provided. $A_{ST} = \frac{3}{4} A_{st}\,(+ve)$

= three quarters of area required for maximum mid span moment in the slab

ek continuous doosra discontinuous me kya condns h? kya amount of Rf denge

Ast(+ve) ky. h?

torsion ko tine layers me deth.

torsion rf top aur bottom dono me kyo deteh?

Ast : 0.75 Ast (+ve) (13.6

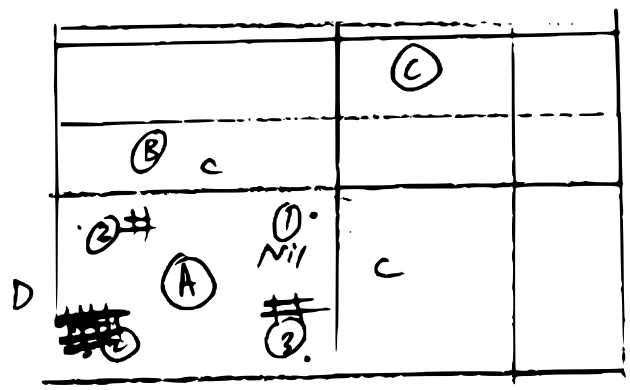
3. When one edge is continuous and other edge is discontinuous Hence torsion reinforcement is provided which is $\frac{1}{2}$ Ast

$= \frac{1}{2} \times \frac{3}{4}$ Ast (+ve)

$= \frac{3}{8}$ Ast(+ve) = 0.375 Ast(+ve)

Ast(+ve) = area of +ve tension rf in slab.

Note: torsion reinforcement is provided in four layers each layers distance is $\frac{l_x}{5}$.



C = continuous

D: discontinuous.

. Torsion reinforcement is provided in top & bottom both side (in 2 layers each = 4 total)

Objective ans

. In case of 2 way slab limiting deflection of slab - primarily function of shorter span.

IES 1997 → . A reinforced concrete slab 75 mm thickness maximum size of reinforcement bar that can be provided $\phi_{max} = \frac{D}{8} = \frac{75}{8} = 9.37$ mm $\simeq 8$ mm.

(roundoff to lower side cuz max)

. Q. 2.11 b. → . In the design of two way slab restrained at all edges torsional reinforcement required is.

Q 2.13 d → → Nil. (restrained matlab continuous edges, continuous edges me Nil AsT ayega).

IS 1995 → Shrinkage deflection in case of rect "g" slabs can be eliminated by putting = Compression steel equal to tension steel

IS 1995 9mp → 22.6 The main reinforcement of an RC slab consists of 10 mm bars at 10 cm spacing. If it is desired to replace the 10 mm bars by 12 cm ~~bars~~ bars, then the spacing of 12 mm bars should be

~~Spacing~~ ~~Area~~

ϕ → S_1

10 mm ϕ → 10 cm

$\frac{\pi}{4} \times 10^2 = A_1$ → $S_1 = 10$ cm

12 mm ϕ → $S_2 = ?$

$\frac{\pi}{4} \times 12^2 = A_2$ → $S_2 = ?$

13.9

[Spacing ∝ Area of bar] or [S ∝ A]

$$\frac{S_1}{S_2} = \frac{A_1}{A_2}$$

$$\frac{10\,cm}{S_2} = \frac{\pi/4 \times 10^2}{\pi/4 \times 12^2}$$

$$= 100\,mm \times \frac{12^2}{10^2} = S_2$$

$$\Rightarrow S_2 = 14.4\,cm \quad \text{(144 mm or)} \leftarrow \text{New spacing}$$

Difference in spacing $= S_2 - S_1$

$= 14.4 - 10$

$= 4.4\,cm.$

IES 1997

Side face is provided when depth of web exceeds → 750

ORS Me primary compression failure kya cause krta h?

flexural collapse in over reinforced beams is due to → primary compression failure

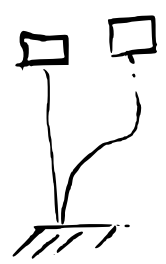
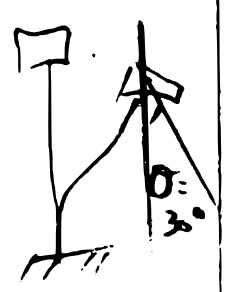
IES 1995

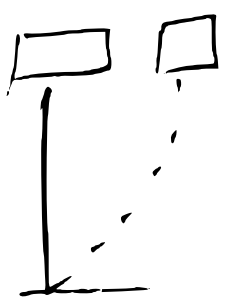
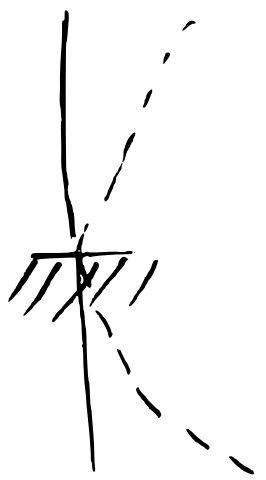
Q from limiting deflection pov use of high strength steel in RC beams results in

→ Increase in depth

Q. Design of one way RC slab for concentrated load is done by → pigeauds moment coefficients.

1 April 2024 Class 14.

Sl no	Degree of end restraint of compression members	figure	Theo eff length	Reco value of eff length
1	effectively held in position & restrained against rotation at both ends		0.50 l or $\frac{l}{2}$	0.65l
2.	effectively held in position at both ends restrained against rotation at one end.		0.70l or $\frac{l}{\sqrt{2}}$	0.80l
3	effectively held in posn at both ends but not restrained against rotation		1.0 l	1.0l
4	Eff. held in position & restrained against rotation at one end, and other end restrd agst rotn but not held in position		1.0l	1.20l
5	effectively held in position & restrained against rotation in one end and at other partially restd against rotation but not held in position.		-	1.5l

	Degree of end restraint of compression members	figure	theoretical effective length	Recommended effective length
6	Effectively held in position at one end but not restrained against rotation and at other restrained against rotation but not held in position		2.0 l	2.0 l
7.	Effectively held in position and restraind in rotation but not held in position not restrained against rotation at other end / Cantilever column		2.0 l	2.0 l

14.3

flat slab ka primary design criteria kya h?

flat slab ka thickness kya hota h?

punching shear kam karne ke liye kya denge?

Drop ya head se kya hota hai?

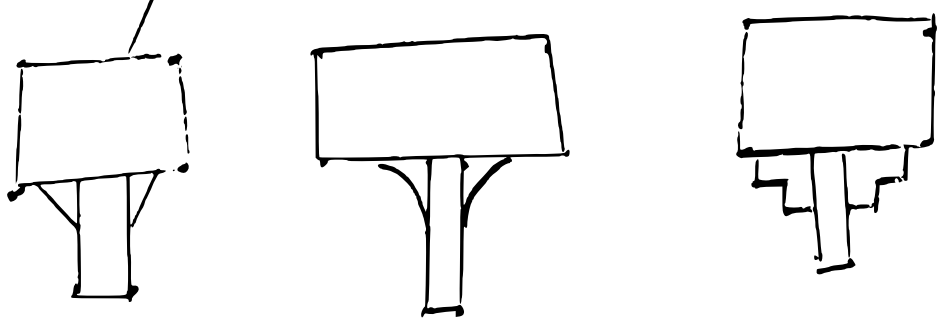
Isotropic slab kya hota h?

Flat slab

Primary design criteria for flat slab is punching shear or two way shear.

thickness of flat slab is ≤ 125 mm

In order to reduce punching shear effect heads or drops are provided.



if drops or heads are provided then area or crossection area increases hence stress decreases

Isotropic slab

Asbtropic slab is defined as the slab ~~for~~ on which longerspan is

Isotropic slab ke examples kya h?

Anistropic slab kisko bolte h?

Thin slab kya hota h?

Yesaare one way h ya two way ya diff category all together h?

Kis base ke under atah thin slab?

equal to shorter span ($l_y : l_x$)

Hence $Ast_{(xx)} = Ast_{(yy)}$

Examples: Square slab, circular slab.

Non isotropic or Anisotropic slab

When $l_x \neq l_y$

$Ast_{xx} \neq Ast_{yy}$

Example rectangular slab

thin slab

thin slab comes under plane stress case. Hence very safe in shear.

effective span of beam or slab.

1. simply supported beam or slab

14.5

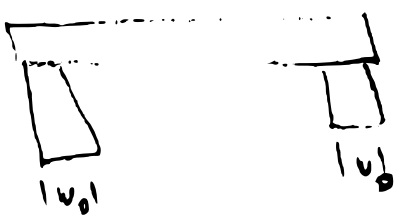
SSB or SS slab ki effective lengths kya hongi?

(a) agar support width l_o ka $\frac{1}{12}^{th}$ ya kumh?

(b) $W > \frac{l_o}{12}$ me?

SSB slab aur fixed slab me kya diff. h?

1.1. when width of support (W) $W \leq \frac{l_o}{12}$



$l_{eff} \neq l_o + d$

$\neq l_o + W$

$\Big\} \downarrow l_{eff}$

agar width $\frac{l_o}{12}$ se kamh toh l_{eff}: (l_o + depth) ya (l_o + width) me se jo kamh wohi hoga.

1.2. $W > \frac{l_o}{12}$

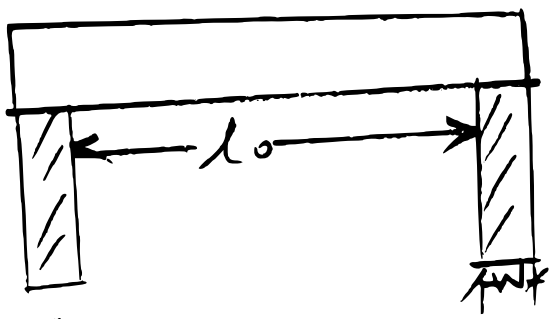
$l_{eff} = l_o$

l_o is unsupported length of beam or slab.

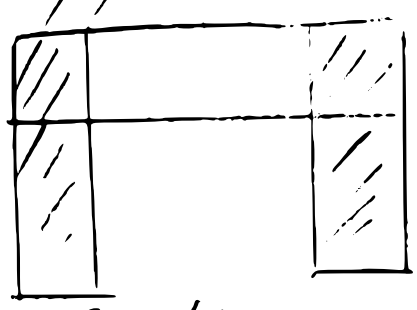
$l_{effective}$ = effective length of beam or slab

d is effective depth of beam or slab.

W is width of support



SSB



Contin fixed beam

(continuous
kab bolenge
slab ko^.
londns kya
h isme
jab w, $\frac{l_o}{12}$
se kam h toh
max l_{eff}
kitni lenge.

Ye simply
supported se
kaise simmilar
h wo $7\frac{l_o}{12}$ wala
condition.
effective length
zyada kyon
le rahe?

kya logich
cantilever
ke l_{eff} ko
$l_o + d/2$ hone
ka?

2. (continuous slab
or beam

(i) when $w \leq \frac{l_o}{12}$

l_{eff} minm $l_o + d$
$l_o + w$
$l_o + 600\,mm$

(ii) when $w > \frac{l_o}{12}$

$l_{eff} \not> l_o$
$\not> 600\,mm$

l_{eff}

(iii) Cantilever beam or slab.

$l_{effective} \not> \left(l_o + \frac{d}{2}\right)$
$\not> \left(l_o + \frac{w}{2}\right)$

14-7 | Lateral stability ke liye slenderness limits kya hongi
kya ye column ke liye applicable hongi?
lateral aur linear deflection me kya farkh?
SSB ke liye lateral stability ke liye deflection limits kisse zyada mai honi chahiye

Cantilever beams/columns ki lateral stability ensure krne ke liye kya condns hz.

Summary

Slenderness limits for beams to ensure lateral stability (also applicable for column)
Pg 39 of code

(a) for simply supported or continuous beam —

$l_o \not> 60\,b$

$\not> 250\,\frac{b^2}{d}$

l_o is clear distance between the lateral restraint (support)
b is compression flange width or simply width.
d is effective depth of beam

(b) for cantilever beam

$l_o \not> 25\,b$

$\not> 100\,\frac{b^2}{d}$

1978 wala
IVSM code
ya old
code h.

IES 1999 ka
97

Q: The bending moment coefficient 1/16
for continuous RC slab in
IS 456:1978 code is based on
Yield line theory.
IS 456:2000 → Westergaards
pattern loading

Q. A doubly reinforced beam is
considered, less economical than a
singly reinforced beam
Options ⓐ Shear reinforcement
is more
ⓑ Concrete is not stressed to its
full value.
ⓒ Tension steel required is more
than the balanced section
ⓓ Compression steel is understressed.

Ans ⓓ Compression steel is
understressed

149

2002 IES Ans -s are important in case of deep beam when compared to flexure alone

(a) axial
(b) shear
(c) torsion
(d) bearing

Ans (b) - Shear

IES 2009 Ans Very important qn, conclusion ko yaad kar lena

In limit state design method the moment of resistance for balanced section using M 20 grade concrete, HYSD grade steel Fe 415 is given by $M_{ulim} = K b d^2$. find out the value of k

Ans: $M_{ulim} = 0.87 f_y A_{st} (d - 0.42 x_{ulim})$

14.10

$x_{u lim}$	Fe
= .53	250
= .48	415
= .46	500
= .44	550

agar M20 grade
bola avo
Fe 415 steel
bola toh
Mulim kitna
gyega Ans d

IES
Q 2008 kq
Q 2011 IES kqan

Shorter span
lx
longer span
ly

$M_{u lim} = 0.36 f_{ck} x_{u lim} (d - 0.42 x_{u lim})$

$= 0.36 \times 20 \times 0.48 \left[1 - 0.42 \times 0.48\right] \times bd^2$

Ans d2
$M_{u lim} = 2.76 bd^2$

Q. for shorter stor.ey = height, cheaper formwork and better lighting facility what is the recommended slab floor → flat slab

for a continuous slab of span 3m x 3.5m size, the minimum overall depth of slab to satisfy vertical deflection limit is (Fe 415 steel is used)

Ans it is a continuous slab wher lx is 3m & ly is 3.5m.

14.11

depth ko span higher side pe round off kiye kyonki max limit 32 h, depth kam hui toh ratio 32 se zada ho jayega jo problem fic h.

$\frac{l_y}{l_x} < 2$ two way slab

$\frac{3.5}{3} < 2$ — " —

	M 35	Fe 415
SS		28 ✓
cont	40	32

$\frac{l}{d} = 32 \Rightarrow \frac{l}{32} = D$

$\frac{l_x}{32} = D = \frac{3000}{32}$ mm $\Rightarrow D = 93.75$ mm

$\simeq 100$ mm.

options (a) 70 (b) 120 (c) 100 (d) 200

Ans (c) 100 mm.

2011 IES ka ☆ important ques

Consider the following statements and state which are correct.
Percentage of steel for balanced design of singly reinforced rectangular section by limit state method depends on -

1. Characteristic strength of concrete (f_{ck})
2. Yield strength of concrete F_y

Asf he formulae mefxulim fckatah 4tah jisme Esatah dafhh foh statement 4 correcth (6johhwoh 6h Astwale me atah) toh stmt 1 correct h. toh stmt 3 correcth.

3. Modulus of elasticity of steel Es (14.12)

4. Geometry of section. b, d, D

Solution: Ast (1.)

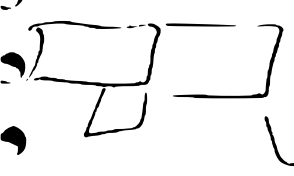
$$0.36 f_{ck} x_{u lim} b = 0.87 f_y A_{st}$$

$$\frac{0.36 f_{ck} x_{u lim} b}{0.87 f_y} = A_{st}$$

$$x_{u lim} = \frac{0.0035 d}{0.0055 + \frac{f_y}{1.15 E_s}}$$

Ans statement 1, 3 & 4 are correct

QIES2006: Centre to Centre distance between beams mafhab do beamsh beech ka distance h



Q A ~~Deep~~ T beam roof section has the following particulars:

(i) thickness of T beam/slab D_f = 100 mm

(ii) Width of web/rib (b_w) = 300 mm

(iii) Depth of beam ~~(bw)~~ = 500 mm

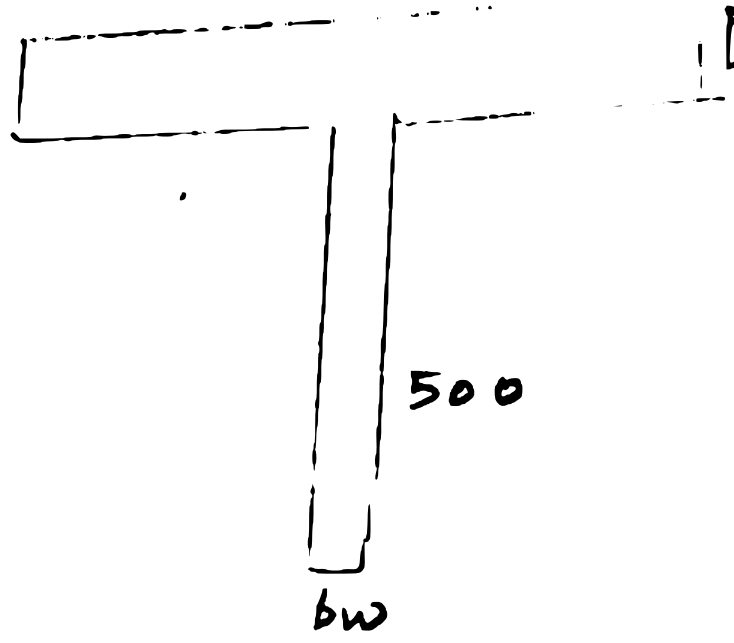
(iv) Effective span of beam l_{eff} = 6 m

(V) Centre to Centre distance of beams = 3 m

(vi) Distance between points of contraflexure = 3.8 m

14.13)

What is the effective flange width of T beam.



Isolated manneke liye flange ka original width nahi diyah.

apan is qn me point of contraflexure ka hi lo maan lenge

Monolithic T beam roof means it is monolithic construction. Monolithic T beam

$$B_f = \frac{l_o}{6} + 6D_f + b_w$$

$\{l_o = 3.6\}$

$$= \frac{3600}{6} + 6\times100 + 300.$$

final selection.

① T formula 1500mm

② ⊥ clc distance between beams 3000mm

③ ⊥ $b_w + \frac{l_1}{2} + \frac{l_2}{2}$ $+ \frac{300}{2} + \frac{3000+300}{2}$

$= 3300mm$ ⇒ Answer = 1500 mm.

apan effective flange width ke liye teeno conditions abc mese minimum wala lenge

(New chapter)
footing aur foundation me kya difference hota h?.
foundation kisme kisme fail hotah?

foundation ke kya kya types hote h?

Deep foundn kya hotah?

Shallow foundn kya hota h?

Examples kya h?

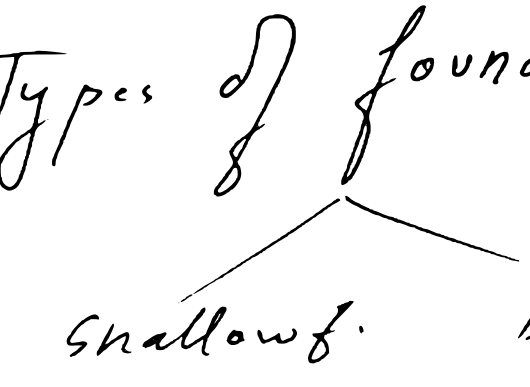
2 April 2024 Class 15

foundation / footing.

foundation: Jab kisi building ke complete base ki baat karte hain ya complete base ko define karte hain toh uch foundation kehlata hai.

footing: Jab kisi column ke base ki baat karte hain individually toh ye footing kehlata hai - foundation fail in bending moment, one way shear (simple shear), two way shear (punching shear)

Types of foundation.



15.2

Shallow foundation me width aur depth ka kya relation hai?

Isolated footings kitne path kyaa examples h?

Combined footing kya hota h?

strip/raft/mat kitna path h?

Raft ya mat foundn kab banega?

pile aur well kiske example h?

Strip aur strap me kya farak hai?

Shallow foundation

$B \geq D_f$

(i) Isolated footings
- → Rectangular
- → trapezoidal
- → circular
- → square

(ii) Combined footings

(iii) strip footing

(iv) Raft or mat footing

(v) Strap footing

Deep foundation

$B < D_f$

(i) Pile foundation

(ii) Well foundation.

When area of column footing is more than 50% of gross area of site. then raft or mat foundation is provided.

Main design criteria of slab (15.3

(i) Bending moment, critical section at face of column.

(ii) Shear force / shear:

→ One way shear: for one way shear critical section present at a distance d from the face of column.

b) two way shear or punching shear: Critical section of two way shear is present at a distance $\frac{d}{2}$ from the face of column in both directions

Diagram

kya main design criteria hai footing ka

Bending moment pe kis jagh pe crit sech miltah

One way Shear pe crit sech kahan mitah?

d/2 from face of column pe crit sech kisko miltah?

Main design criteria of footing | 15.3

(i) Bending moment - for bending moment, critical section is face of column.

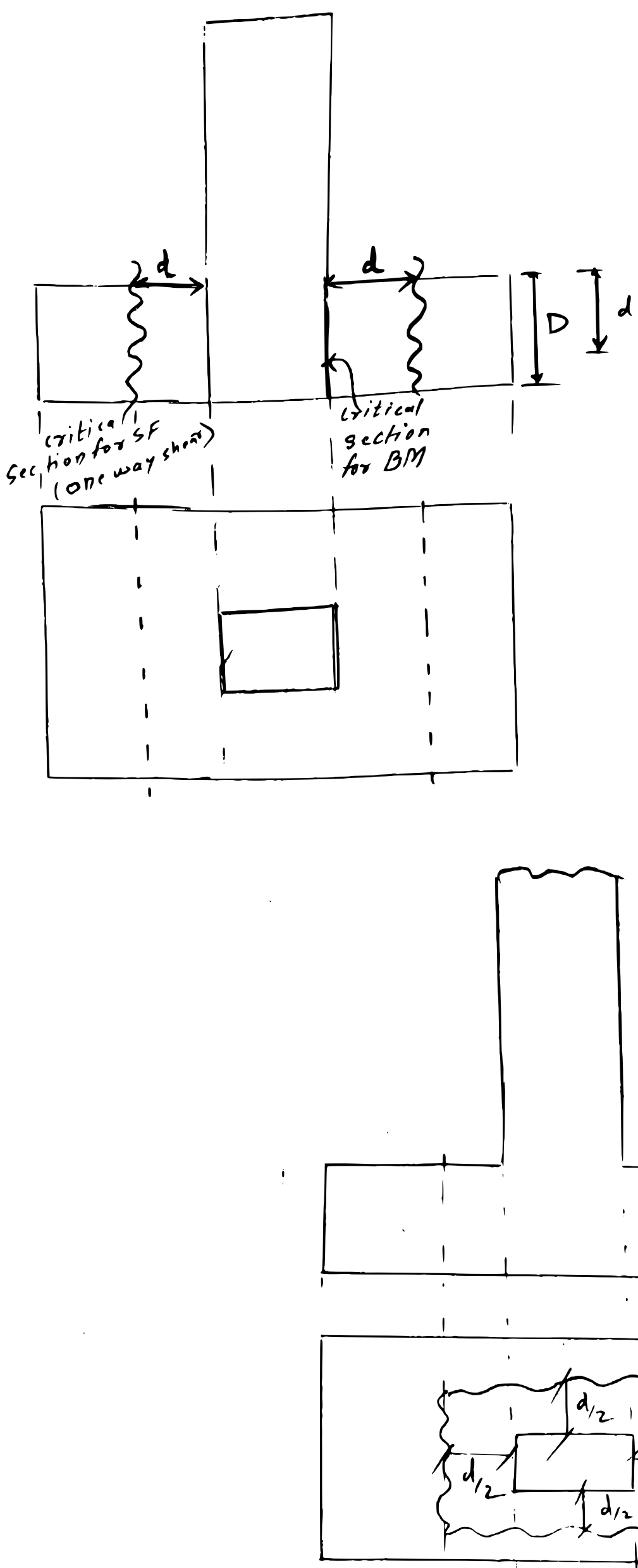
(ii) Shear force / Shear:

⇒ One way shear: for one way shear critical section is present at a distance d from the face of column.

⇒ two way shear or punching shear: Critical section of two way shear is present at a distance $\frac{d}{2}$ from the face of column in both directions

Diagram →

15.9


d
d
D
d
critical section for SF (one way shear)
critical section for BM
d/2
d/2
d/2
d/2

RCC me edge me thickness kitne se kam hai hona chahiye pile he top pe footing me min edge thickness kya hogi.

PCC pedestal ke liye tan α kya hoga? α kya angle h? Imp formula yc valid kiske liye h? q0 kya h?

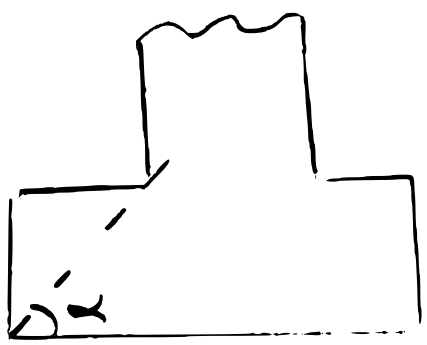
In RCC and plain concrete footing the thickness at the edge shall not be less than 150 mm (for footings on soil)

For footing on top of piles, it should not be ~~more~~ less than 300 mm.

for PCC pedestal

$$\tan\alpha \leq 0.9\sqrt{\frac{100 q_0}{f_{ck}} + 1}$$

α is angle between the plane passing through bottom edge of pedestal & corresponding junction edge of column with pedestal and horizontal plane.



f_{ck} = characteristic compressive strength of concrete at 28 days in N/mm^2.

q_0 is calculated max bearing pressure at the base of pedestal in N/mm^2

"WSM"
permissible
bearing stress
WSM
&
Bearing stress
LSM me kya
hoga

Punching shear
ka formula
WSM me
kya hoga?
LSM me kya
hoga?

K kise kyun
mai thu ek/a2
diya formula
ka.

Projection kyun
liya ha?

Bearing stress - formula

"WSM"

Permissible bearing stress

$q_a = 0.25 f_{ck}$

(ii) LSM : Bearing stress

$q_a = 0.45 f_{ck}$.

Punching shear :

formula : WSM

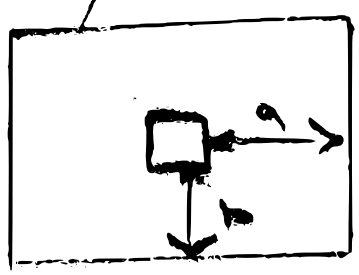
$\tau_{vp} = 0.16 \sqrt{f_{ck}} * k$

In LSM

$\tau_{vp} = 0.25 \sqrt{f_{ck}} * k$

$k = (0.5 + \frac{b}{a}) \not> 1$

a, b : projections in both directions



15.7

RCC column ke liye crit Sect kahan hoga?

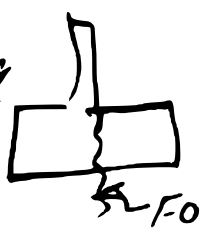
Masonry ke liye crit section kahan hoga?

agar gusset base ka crit section BM kya lenge?

Critical section for bending moment

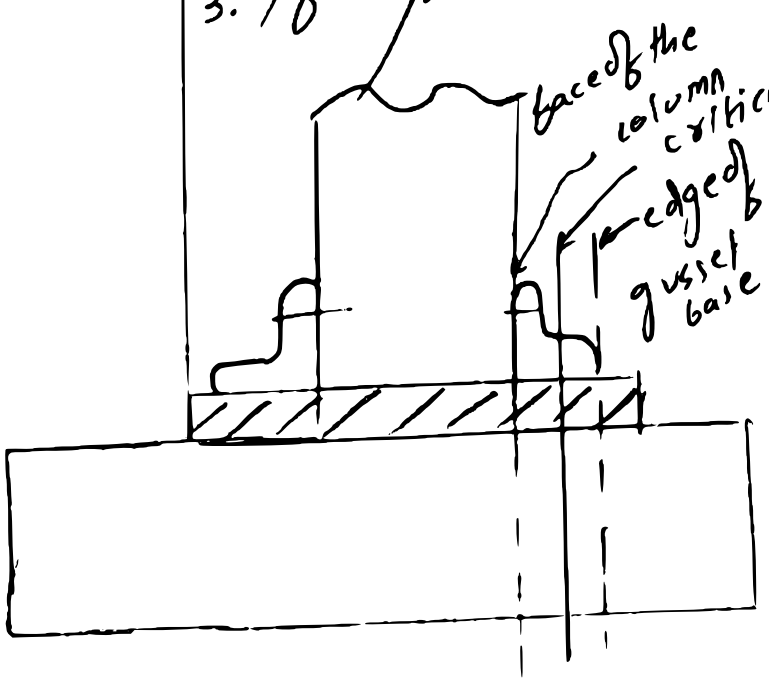
For RCC column or wall or pedestal -

1. critical section for bending moment is face of column.



2. for masonry (or brickwork) column, wall, pedestal - critical section for bending moment is halfway between the centre line and edge of the wall.

3. If gusseted base is provided critical section for BM is halfway between the face of column or pedestal and the edge of gusseted base for footings under gusseted base.



15 .2)

combined footing ko design krte time net soil pressure kaisa milna chahiye? kaise ensure karenge yeh?

JES 2008 kaun

While designing combined footing the resultant of column load passes through the centre of gravity of footing slab such that the net soil pressure obtained is - uniform

→ How is the depth of isolated column footing designed. or governed -

by max BM

by max sf (one way shear)

by max two way shear or punching shear

Jab combined footing design kr rhe ho toh . column ka result, centre of gravity slab se ~~kaise~~ pass kare toh soil pressure kaisa milega?

RCC braced walls kaise design karenge?

Uski min thickness kitni hogi?

Wall me maximum effective height to thickness ratio zyada se zyada kitna hoga?

Walls me reinforcement kaise denge.

wall me kitna hogga?

Vert ref / grossarea conc

WALLS

for braced RCC walls subjected to vertical compression only may be designed by empirical formulae.

The minimum thickness of RCC wall shall not be less than 100 mm (it can be more than or equal to 100 mm.)

The ratio of effective height to thickness of wall shall not exceed 30.

$$\frac{H_{we}}{t_w} \not> 30$$

Minimum requirement for reinforcement in walls

1) The reinforcement for walls shall be provided as below.

a) The minimum ratio of vertical reinforcement to gross area of concrete $\left(\frac{A_v}{A_g}\right)_{min}$ shall not be less than 0.0012 for deformed bars (max dia of bar upto 16 mm)

15.10

Doosre type ke bars ke liye $\frac{A_v}{A_{g\,min}}$ kam Secam kitna hoga?

Welded wire fabric ke liye $\frac{A_v}{A_g}$ kitna hoga? upto what dia?

$\left(\frac{A_v}{A_g}\right)_{min}$ * 0.0015 for other types of bar

$\left(\frac{A_v}{A_g}\right)_{min}$ * 0.0012 for welded wire fabric (max diameter of bar upto 16 mm)

$\left(\frac{A_v}{A_g}\right)_{min}$ * 0.0012 HYSD Fe 415 & more $\phi_{max} \leq 16$ mm

* 0.0015 (other bar)

* 0.0012 [$\phi_{max} \leq 16$ mm] welded wf.

(b) maximum spacing of Vertical reinforcement should not be more than 3 times of thickness of wall or 450 mm whichever is less

$S_{max} \neq 3t_w$

450 mm

wall
Vert spacing of me ~~max~~ zyadase zyada kitna space kasenge?

:: kam se kam ratio kya hogg horizontal of to gross concrete areq?

(a) HYSD kelie

(b) MS kelie

(c) Welded wire fabric ke liena?

Zyadase Zyada horiz of kitna doon rakhsktPh

kab of ko dogrid medenge Wall me 2 kahan 2 denge? kitne thickness?

© Minimum ratio of horizontal reinforcement to gross concrete area

HYSD

$\frac{A_{sh}}{A_{g\,min}}$ ✗ (i) 0.0020 for bars Fe 415 & above

$\phi_{max} \leq 16$ mm

✗ (ii) 0.0025 for other types of bar. (MS Steel)

✗ (iii) 0.0020 for welded wire fabric

[$\phi_{max} \leq 16$mm]

(d) maximum spacing of horizontal reinforcement $S_{max,4}$ ✗ 3tw ✗ 450mm

Note:

for walls having thickness more than 200mm vertical & horizontal reinforcement shall be provided in two grids one near each face of the wall.